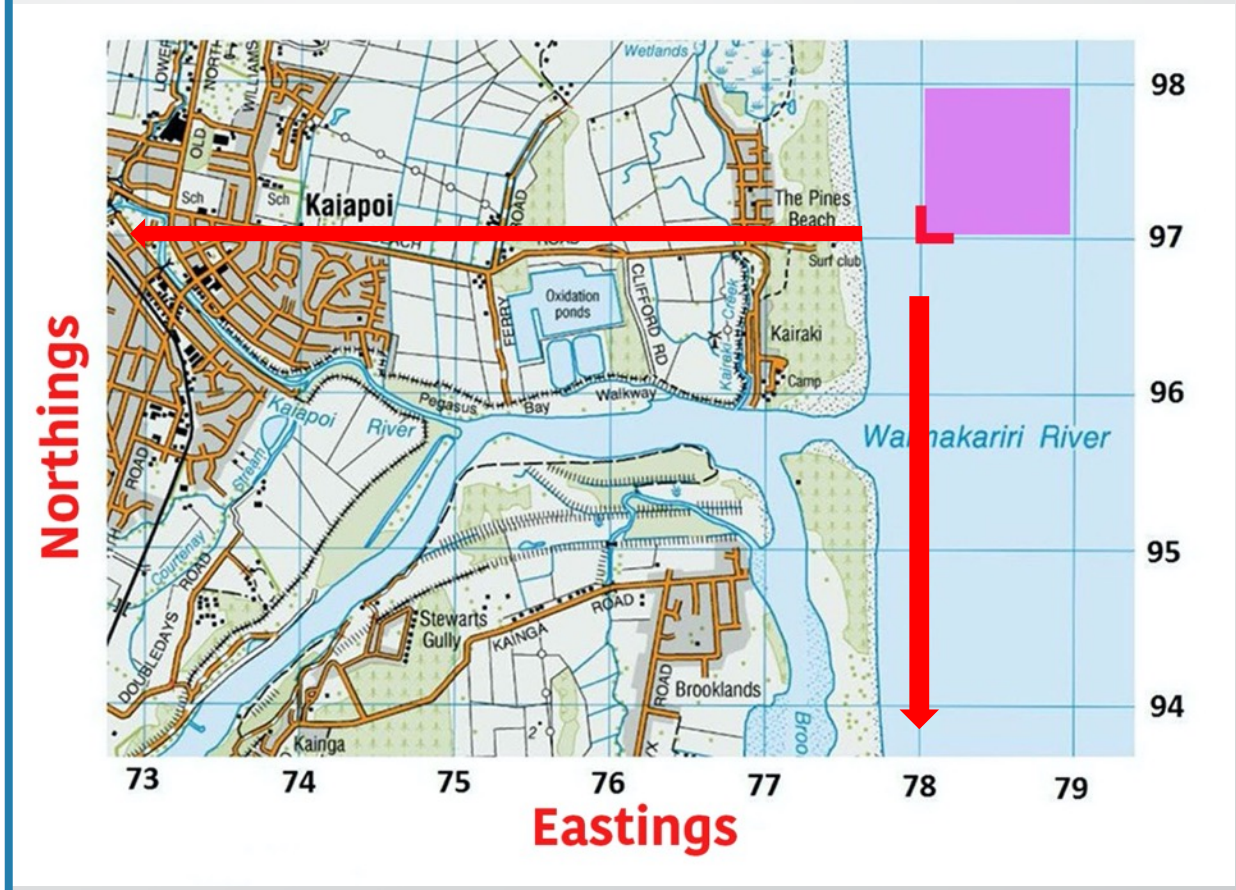
The A-Z of Geo Skills

Year 8

A Area Reference

The A-Z of Geo Skills

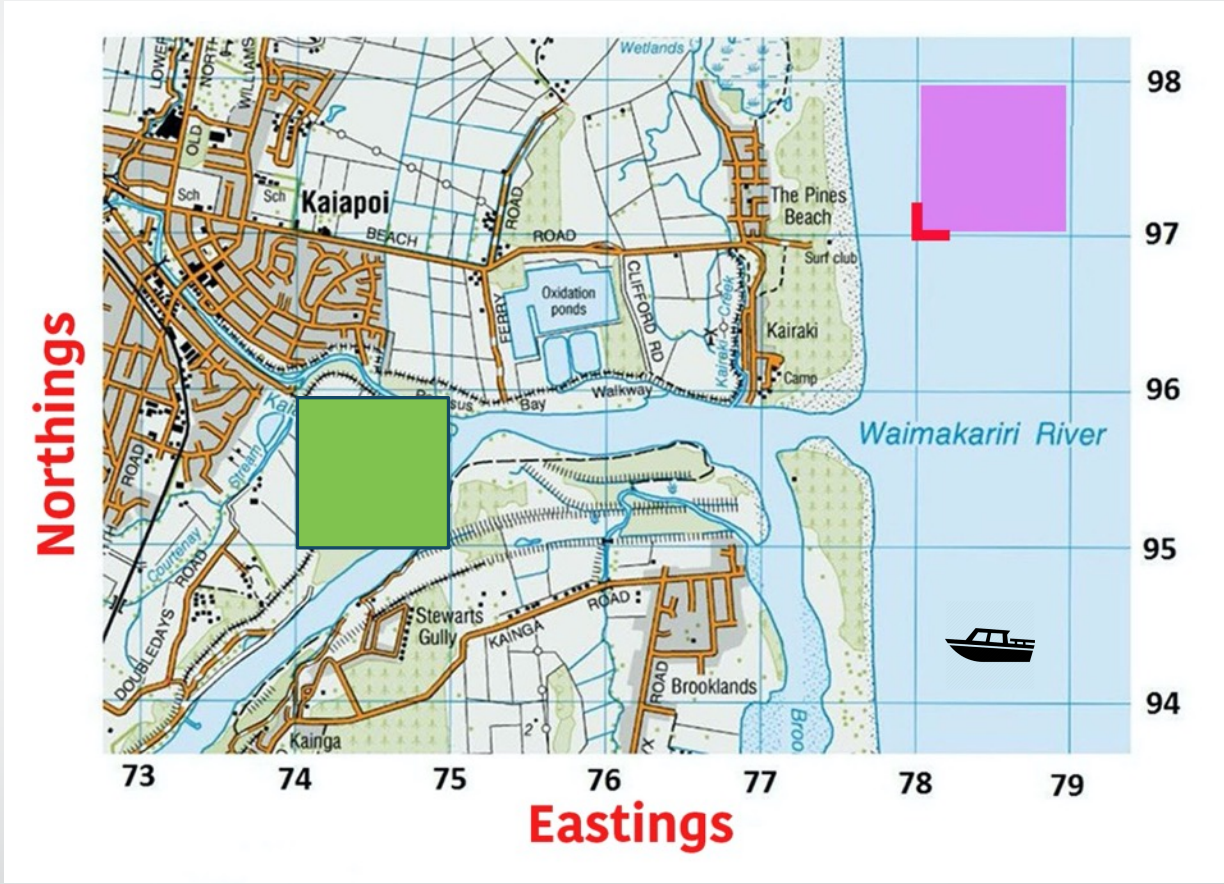
- A sequence of 4 numbers used to locate relatively large features on a topographic map
- Always put <u>eastings</u> first followed by <u>northings</u>
- Not to be confused with Grid References which have 6 numbers
- Abbreviated as AR
- **TIP:** Go the bottom left-hand corner of a grid square to get the numbers you require



The Area Reference of the Pink Square above is **AR 7897**.

A

Area Reference



The Area Reference of the Pink Square above is **AR 7897**.

EXAM STYLE Q

Q1. What is the Area Reference of the green square? Answer: AR 7495

Q2. In what Area Reference is the boat? Answer: AR 7894

The A-Z of Geo Skills

Area Reference

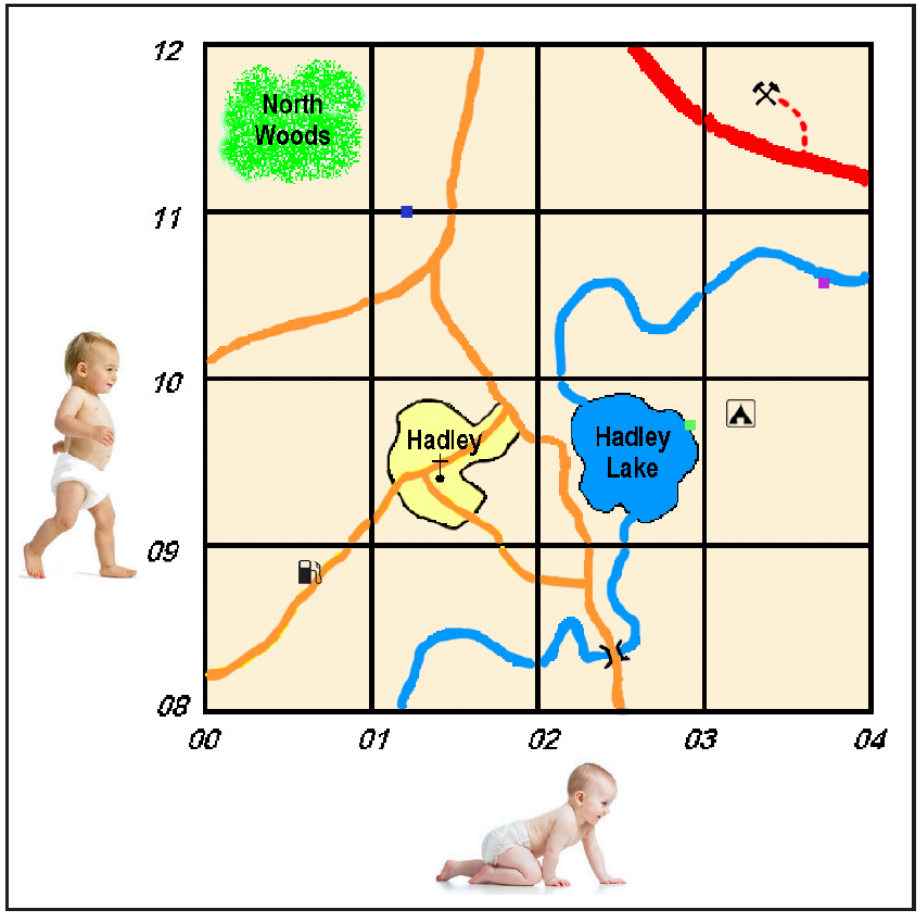
3. AREA REFERENCE

- abbreviated as AR, has 4 numerals
- used to locate large features on maps eg, lakes, forests
- must read eastings figure before northings
- go to the bottom left hand corner of the grid square to calculate
- **TIP: crawl before you can walk**

Practice Q: What is the AR of Hadley Lake

Go to the bottom left hand corner, take eastings reading, then northings.

Answer: Area Reference of Hadley Lake = AR 0209



Here is a quick summary

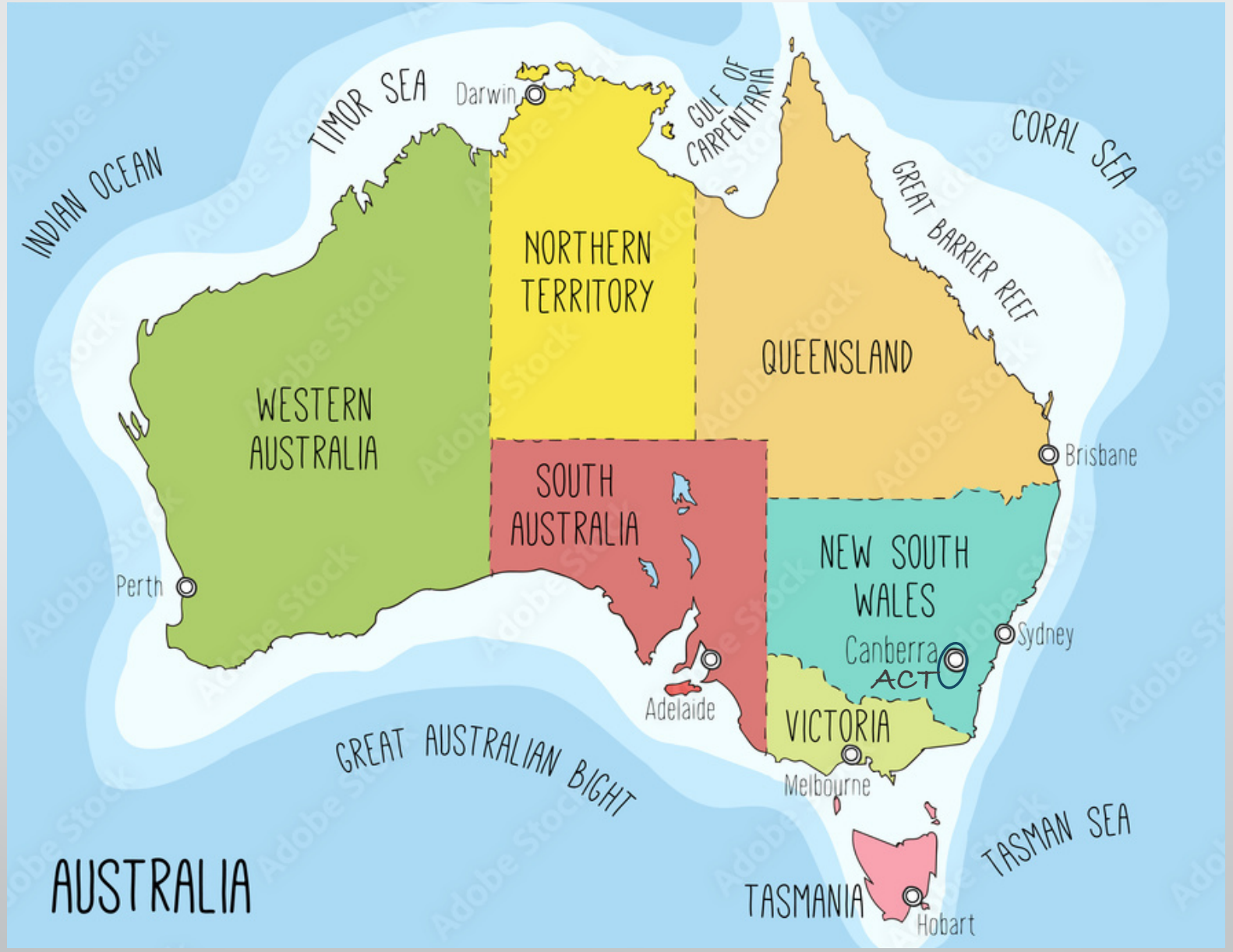
A

Aussie Place Geo

The A-Z of Geo Skills

- Hopefully you learned your States (6), Territories (2) and Capital Cities in Primary School!

- If not, here is a quick refresher.



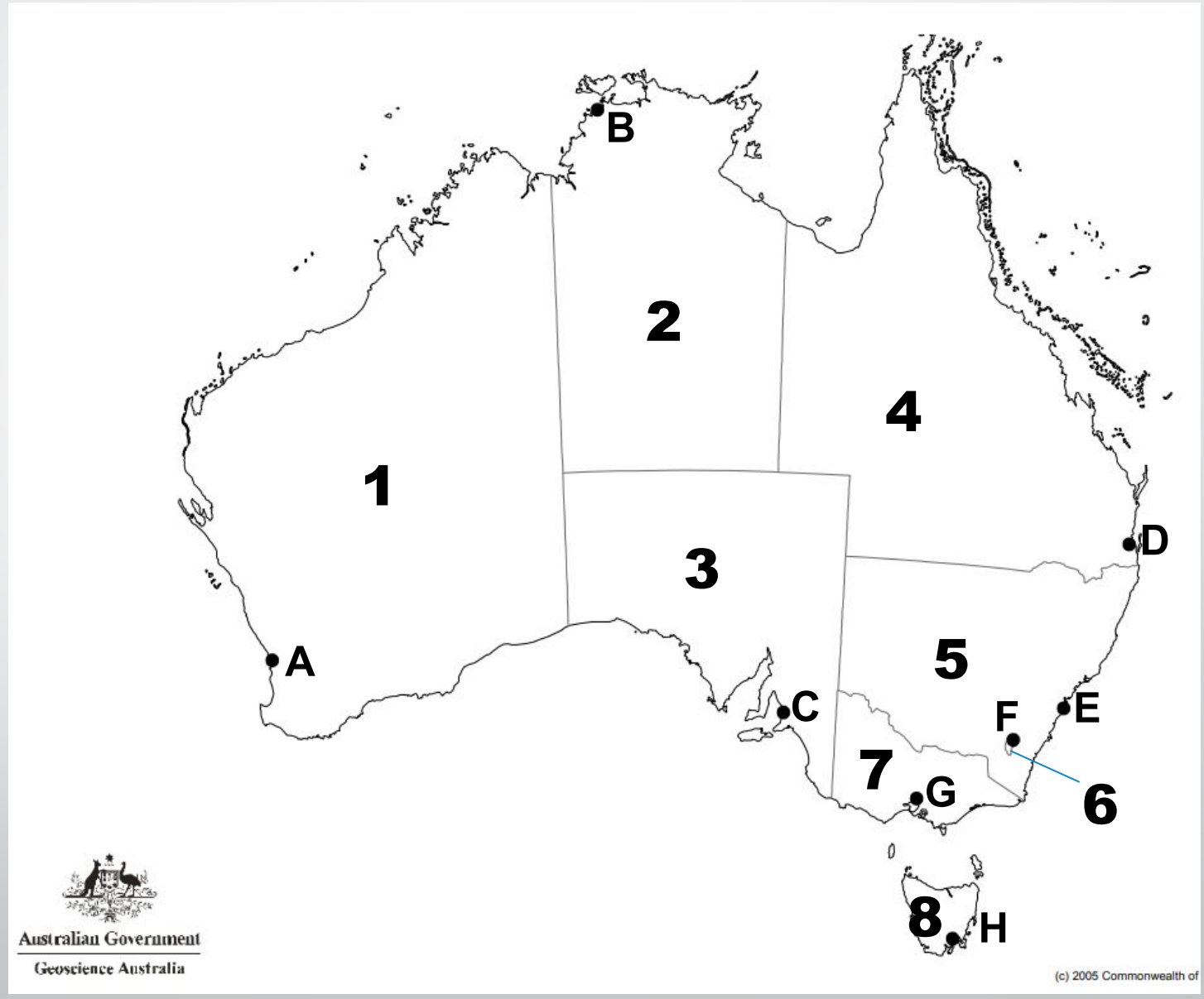
A

Aussie Place Geo

The A-Z of Geo Skills

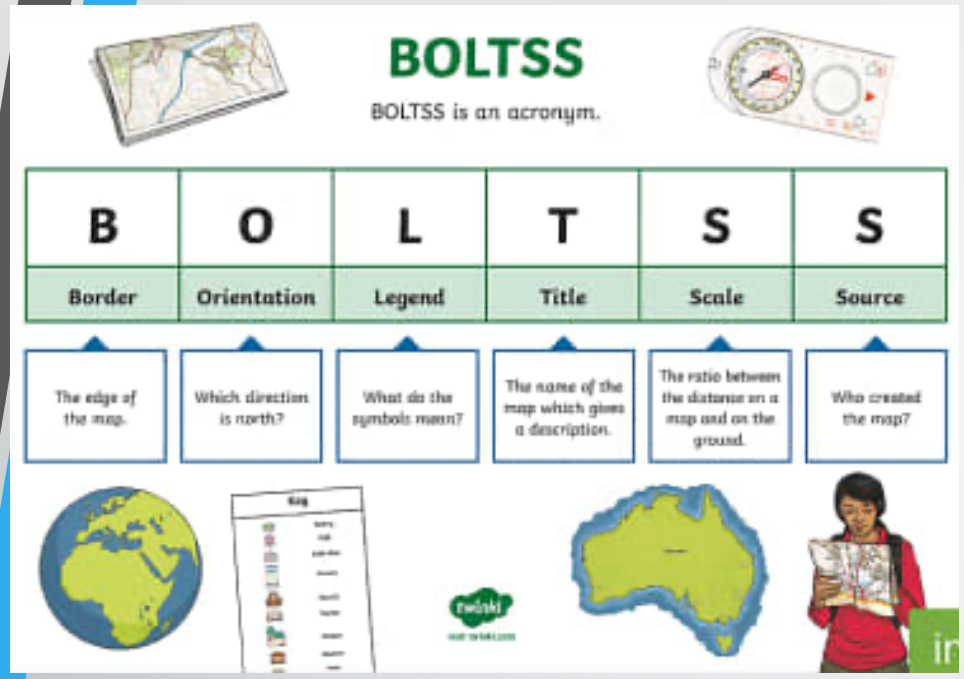
EXAM STYLE Qs

1. What capital city is “A”?
   Perth
2. What capital city is “G”
   Melbourne
3. What capital city is “H”?
   Hobart
4. What State/Territory is “2”?
   NT
5. What State/Territory is “3”?
   SA
6. What State/Territory is “6”?
   ACT



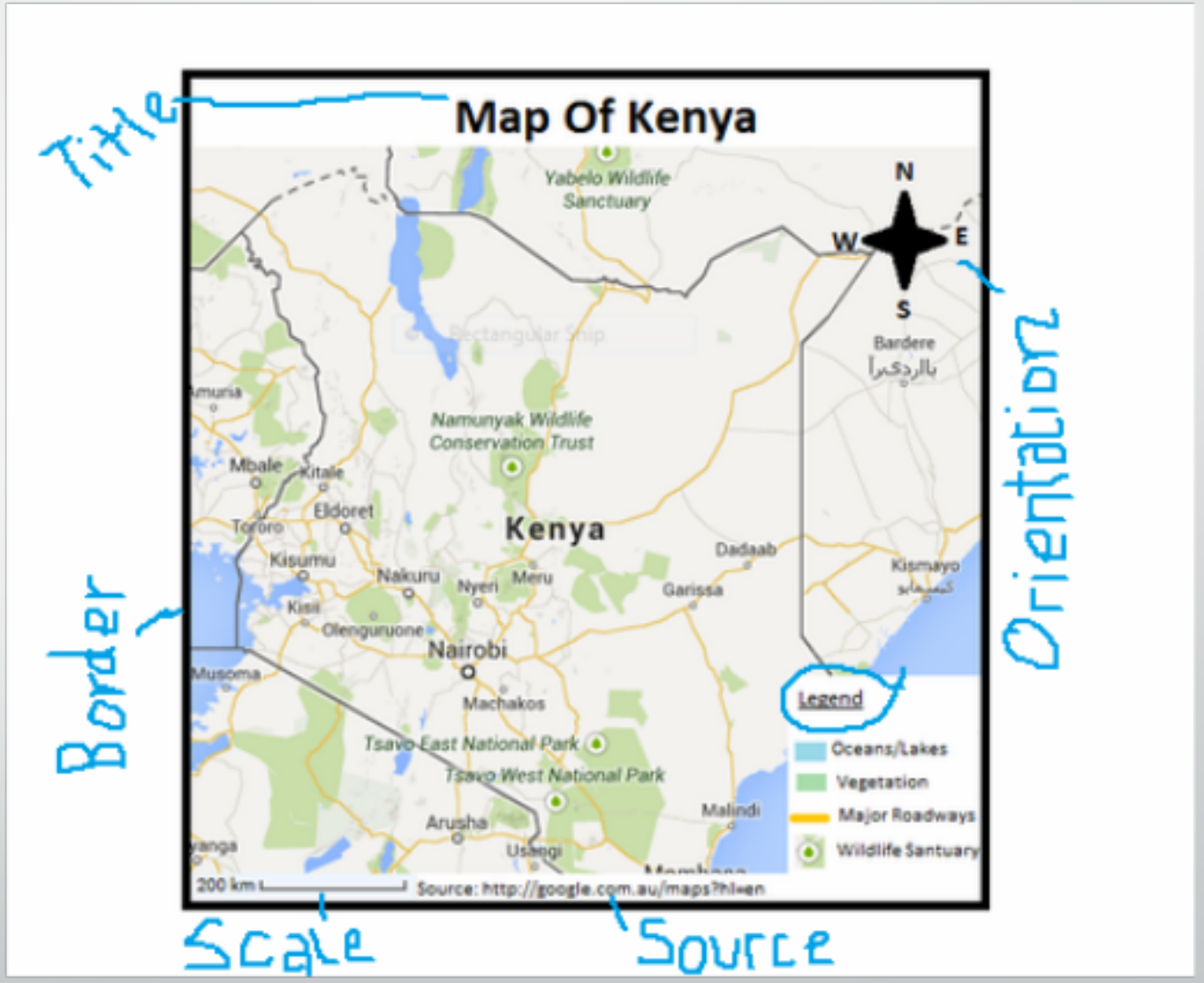
B

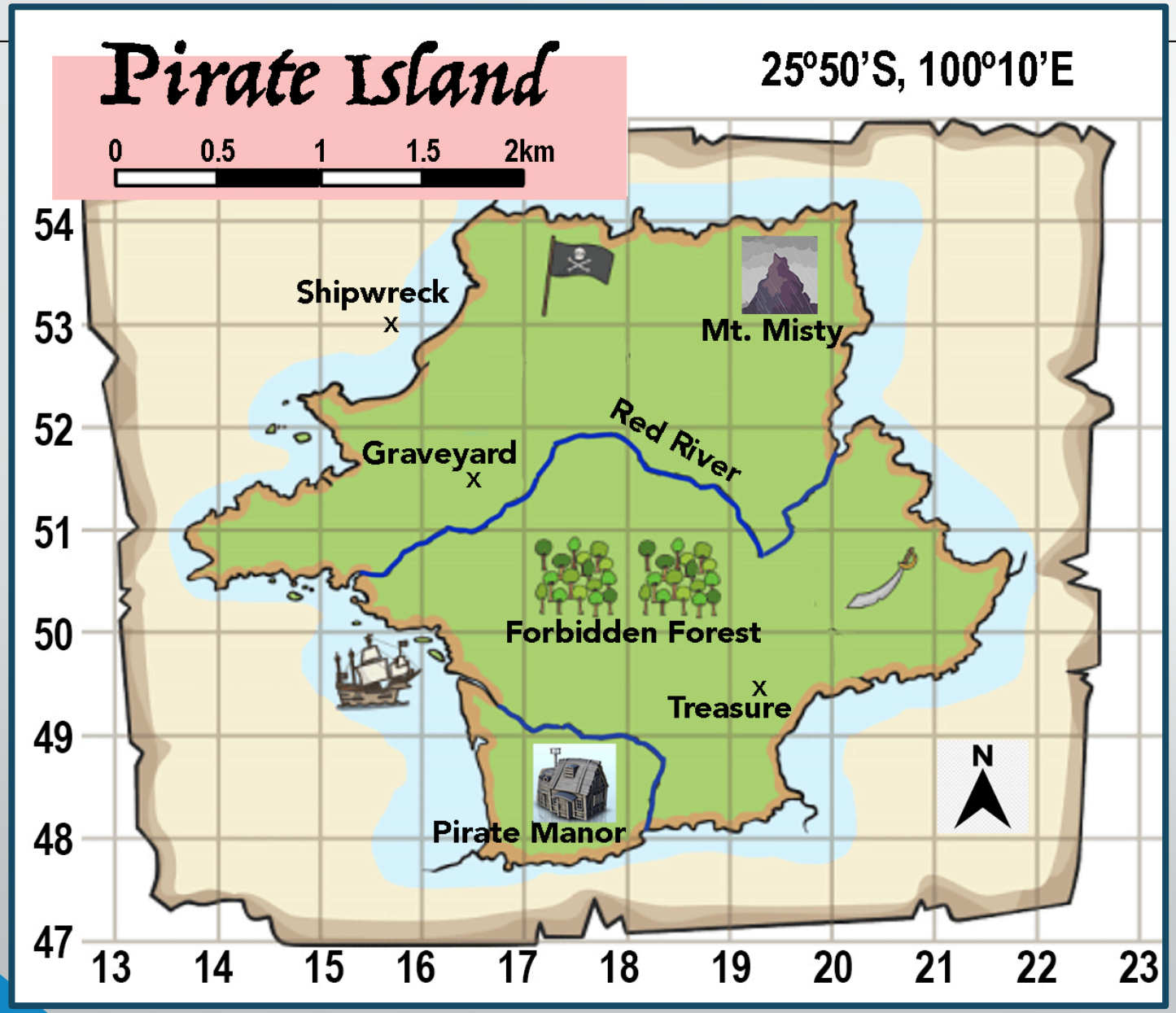
BOLTSS



BOLTSS stands for:

Border
Orientation
Legend
Title
Scale
Source





Exam Style Q:

1. What element of BOLTSS is located in AR 2148?

Answer: Orientation

2. What 2 elements of BOLTSS are missing on the Pirate Island Map?

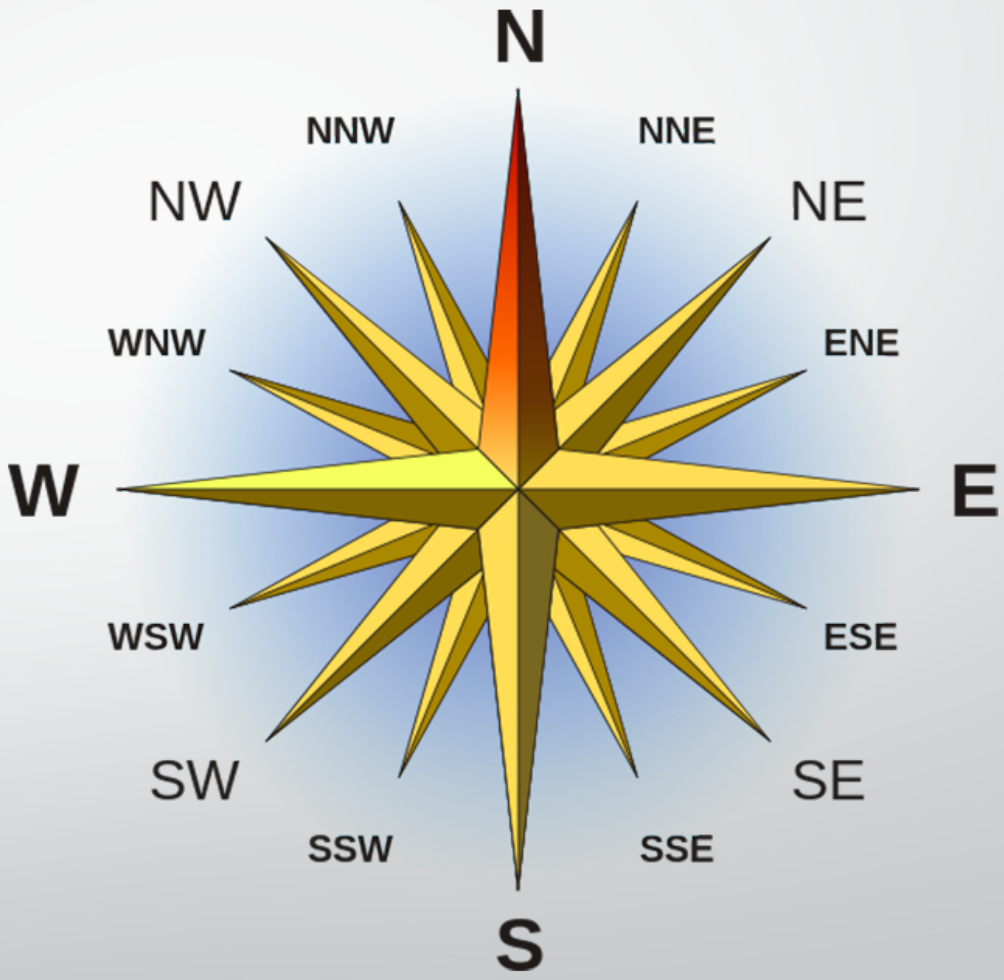
Answer: Legend + Source

D Direction

The A-Z of Geo Skills

- Direction is important because it shows where one place is in relation to another
- North, South, East & West are known as the <u>cardinal points</u> of a compass while the others are known as <u>intermediate points</u>

- **TIPS:** <u>N</u>ever <u>E</u>at <u>S</u>oggy <u>W</u>eetBix will help you remember the 4 cardinal points; you need to learn NE, SE, SW & NW; if you forget the others then always go to the cardinal point first ~ eg. the direction between N & NW is <u>N</u>NW



D Direction



ANSWERS:

Q1) What direction is Armidale from Newcastle?
N

Q2) What direction is Dubbo from Tamworth?
SW

Q3) What direction is Broken Hill from Albury?
NW

Q4) What direction is Newcastle from Lismore?
SSW

Q5) Which region borders Northern NSW to the south?
Mid North Coast

Q6) In what direction is Port Macquarie from Sydney?
A) N
B) NE
C) NNE
D) ENE

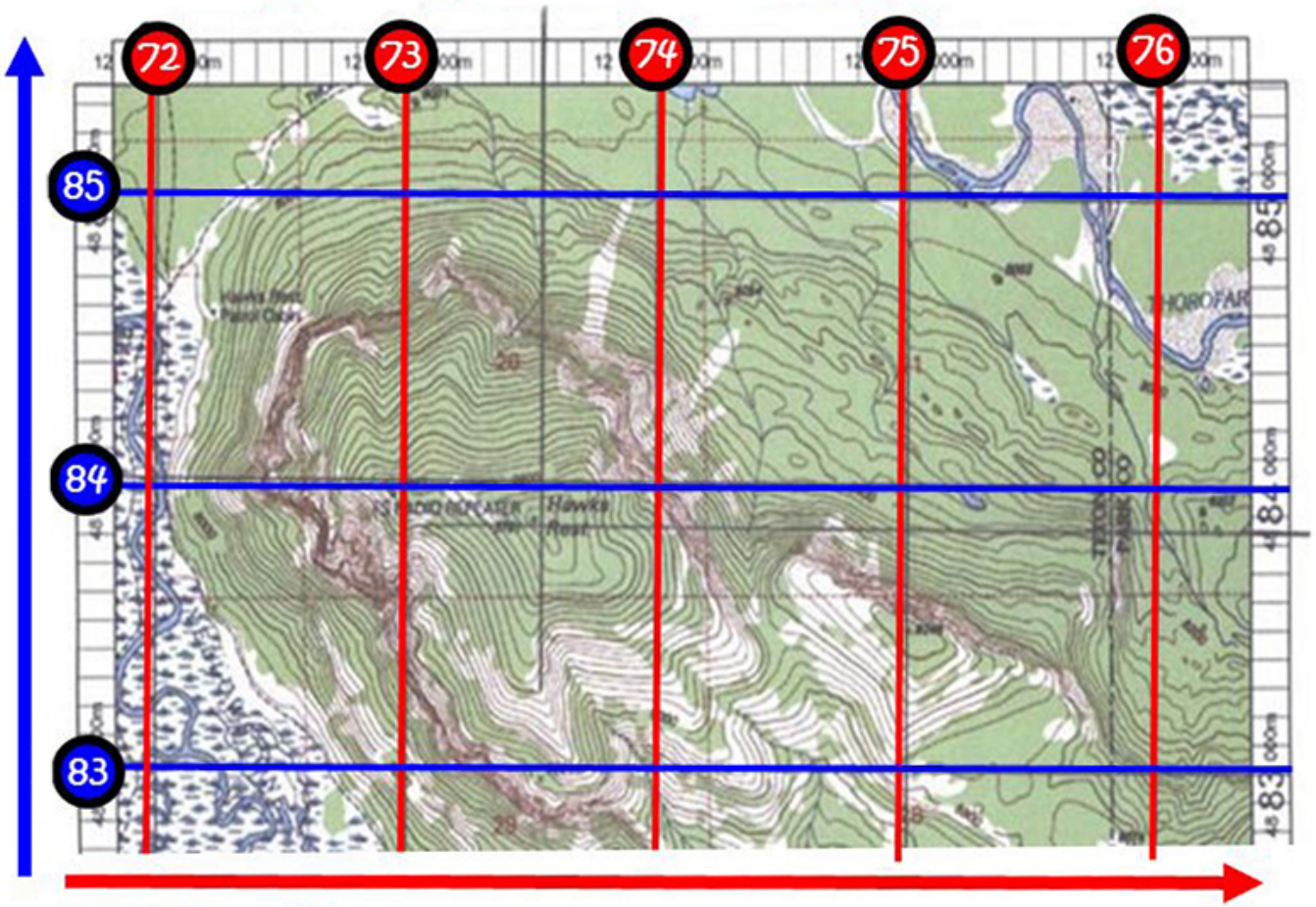
E

Eastings & Northings

The A-Z of Geo Skills

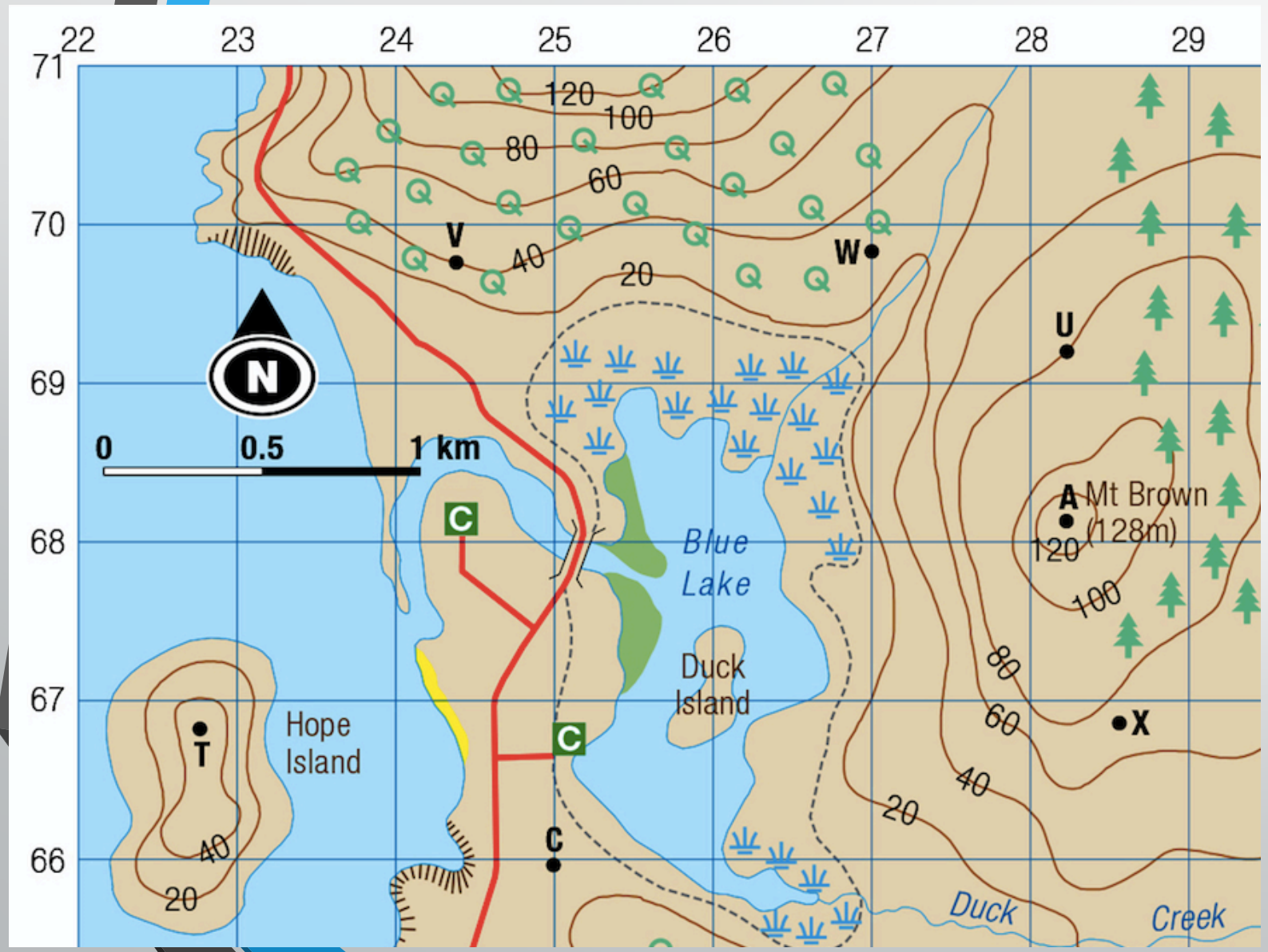
- **TIP:** Eastings always come before Northings (‘E’ comes before ‘N’ in the alphabet might be a good way to remember)

Northings are horizontal grid lines that increase from South to North



Eastings are vertical grid lines that increase from West to East

Eastings & Northings



EXAM STYLE Qs

Q1: What is the closest northing to Mount Brown?

Answer: 68

Q2: Which easting runs through Duck Island?

Answer: 26

Q3: Which northing is closest to Point X?

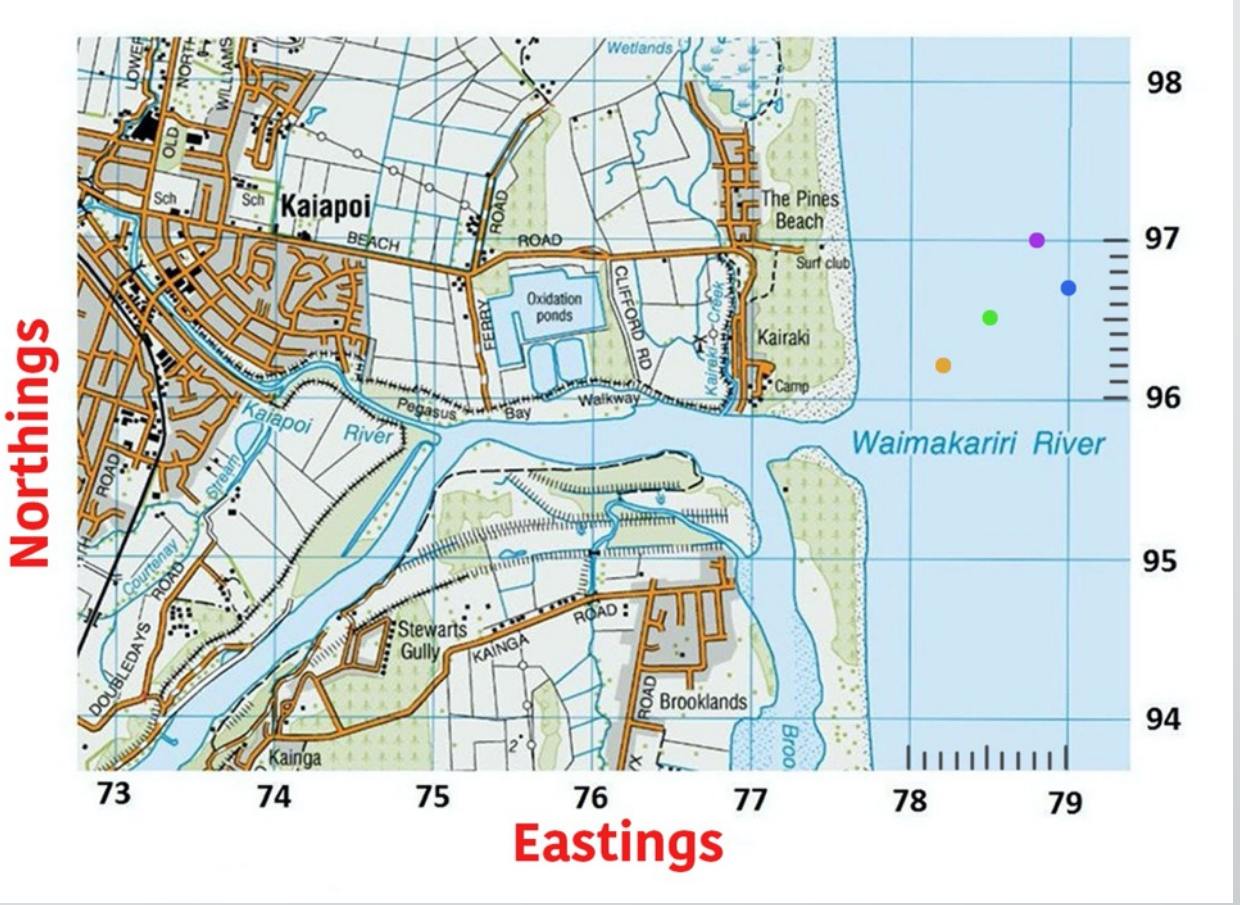
Answer: 67

G

Grid Reference

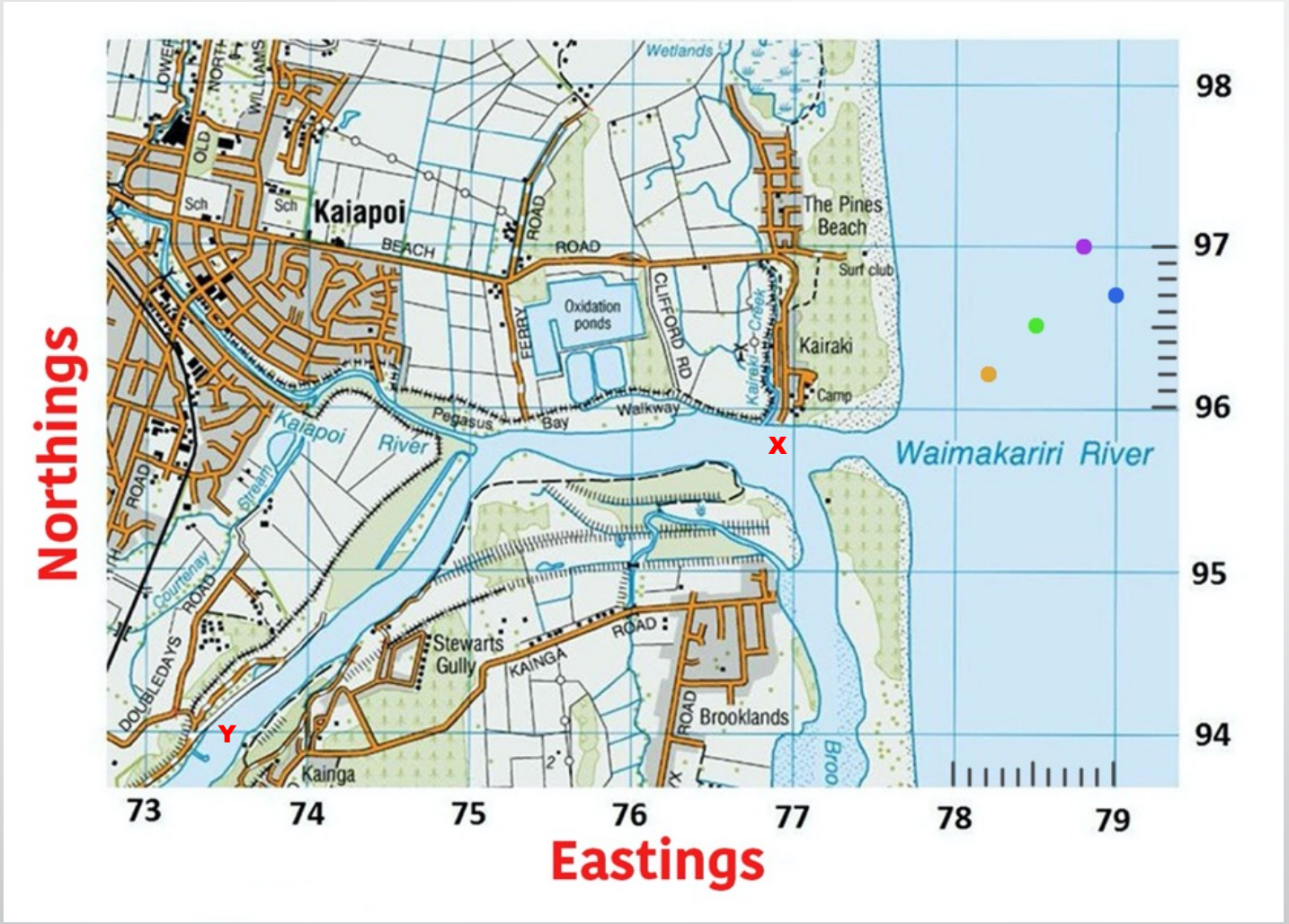
The A-Z of Geo Skills

- A sequence of 6 numbers used to locate features or a precise point on a topographic map
- Always put 3 eastings numbers first followed by 3 northings numbers
- Not to be confused with Area References which have 4 numbers
- Abbreviated as GR
- **TIP:** visualise the numbers as you would those on a ruler with each number separated by 10 intervals



The GR of the orange dot is **GR 782 962**.
The GR of the green dot is **GR 785 965**.
The GR of the purple dot is **GR 788 970**.
The GR of the blue dot is **GR 790 967**.

Grid Reference



EXAM STYLE Q

Q1. What is the GR of location **X**? **Answer: GR 769 958**

Q2. What is the GR of location **Y**? **Answer: GR 735 940**

N

Human v Natural

The A-Z of Geo Skills

- Geographers need to classify things they see into different groups

- Natural features (aka physical features) are those things that are evident due to nature

- Natural features may include rivers, lakes, mountains, cliffs, rocks, soil, vegetation and native fauna

- Human features (aka cultural features) are those that have been made by humans and include roads, buildings, bridges, railways and walking/hiking trails



QUESTION: See if you can list TWO natural and TWO human features evident in the ground level photograph of Mt Taranaki, NZ above.

N

Human v Natural



ANSWER:

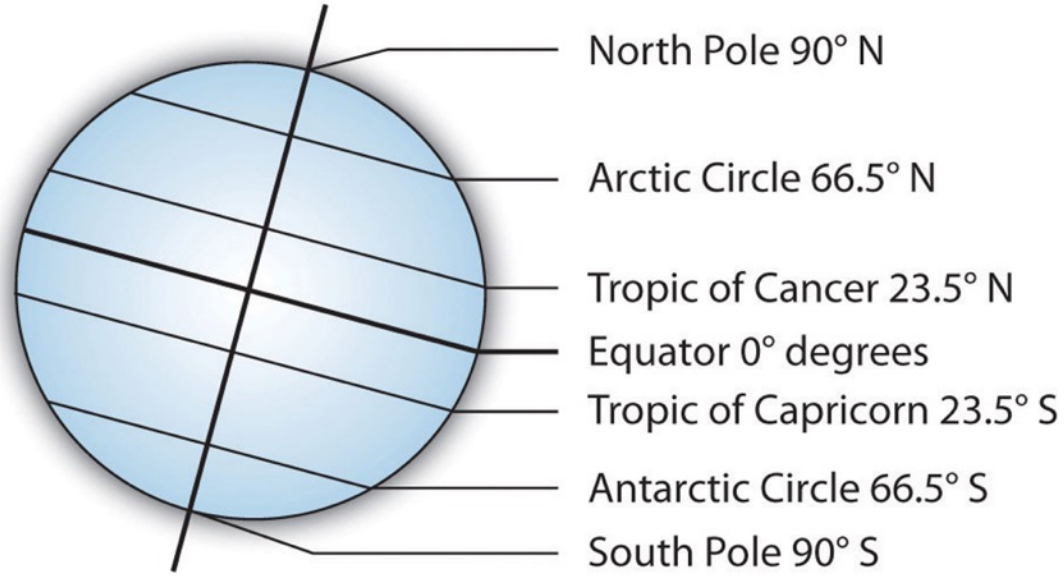
NATURAL or PHYSICAL: **Mountain, trees**

HUMAN or CULTURAL: **Road, fence**

L Latitude

The A-Z of Geo Skills

- Lines of latitude are imaginary horizontal lines that run from east to west
- They are also called parallels of latitude as they are parallel to the Equator (which is 40,075 km in length)
- The maximum latitude is 90^0
- The latitude of a place is crucial in determining climate
- Equatorial places are warmer because they are closer to the sun (the poles are furthest)
- There are 5 major lines of latitude (shown adjacent)
- **TIP:** LAT is FLAT!



ANSWERS:

Q1) What is the maximum latitude?
90^0

Q2) Which Tropic runs through Australia?
Tropic of Capricorn

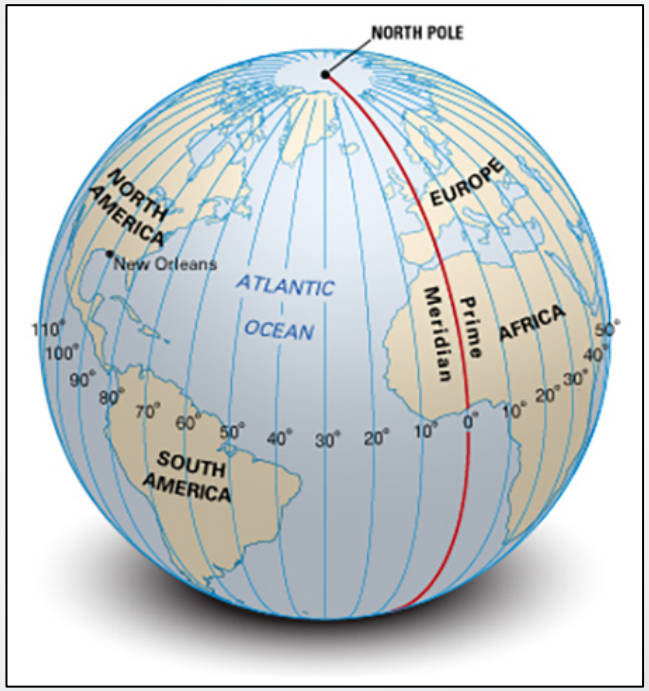
Q3) True or False, all lines of latitude are the same size?
False, they get smaller as you approach the poles

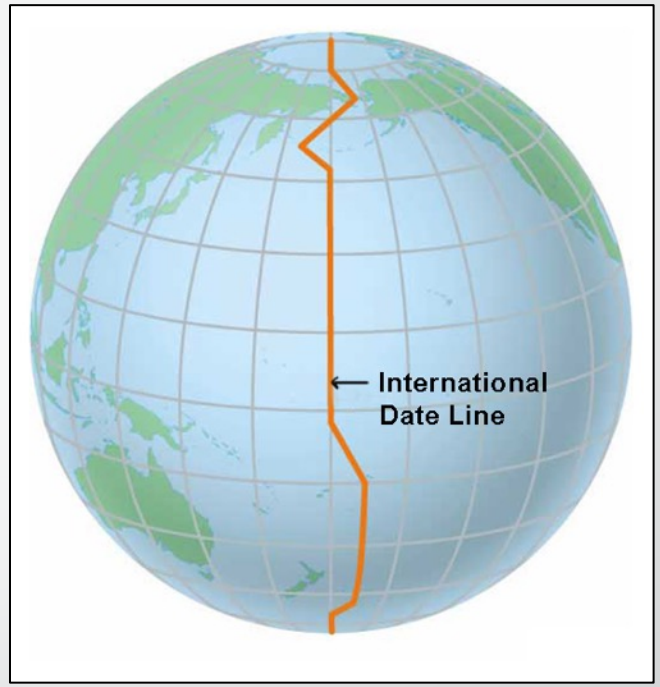
L Longitude

The A-Z of Geo Skills

- Lines of longitude are imaginary vertical lines that run from north to south
- They are also called meridians
- They are measured east or west of the Prime Meridian aka the Greenwich Meridian after the London suburb through which it runs
- They are farthest apart at the Equator and meet at the poles
- The max longitude is 180^0
- There are 2 major lines of longitude (shown adjacent)

- **TIP:** LONG IS LONG!





ANSWERS:

Q1) What is the maximum longitude?
180^0

Q2) Would New Orleans and New York be E or W of Greenwich?
West

Q3) Why is the International Date Line crooked in parts?
So that countries don't have towns/cities where there are separate days occurring a few metres apart!

L Latitude & Longitude

The A-Z of Geo Skills

- Latitude is ALWAYS written before longitude

- In the map adjacent, Rio de Janeiro's location is 23ºS, 43ºW…. while Porto Alegra's location is 30ºS, 51ºW.

- **TIP:** LAT is before LONG in the dictionary … and so it is when you give lats and longs



Latitude & Longitude

The A-Z of Geo Skills



Q1:

What is the location of Sao Luis? (using lat + long)

A. 2°S, 44°E

B. 8°S, 44°W

C. 2°S, 44°W

D. 2°S, 54°W

Answer:

C: 2°S, 44°W

Q2:

What is the location of Boa Vista? (using lat + long)

A. 2.5°S, 60.5°W

B. 2.5°N, 60.5°W

C. 2.5°S, 59°W

D. 2.5°N, 59°E

Answer:

B: 2.5°N, 60.5°W. **It's north of the equator!**

Map quadrants

The A-Z of Geo Skills

As the name suggests, there are 4 map quadrants:

- NW
- SW
- NE
- SE



Map quadrants



EXAM STYLE Q

In what map quadrant is Australia? **Answer: South East (SE)**

World Map

Learn your continents; oceans and major mountain ranges



The A-Z of Geo Skills


Major Mountain Ranges of The World
SCANDINAVIAN
Galdhøpiggen
(2469 m)
Mount Narodnaya
(1894 m)
URALS
Mont Blanc
(4809 m)
ALPS
Belukha Mountain
(4506 m)
ALTAI
ROCKIES
APPALACHIANS
Mount Elbert
(4401 m)
Mount Mitchell
(2037 m)
Mount Toubkal
(4167 m)
ATLAS
Mount Everest
(8849 m)
HIMALAYAS
Equator
ANDES
Mount Aconcagua
(6961 m)
DRAKENSBERG
Thabana Ntlenyana
(3482 m)
GREAT
DIVIDING RANGE
Mount Kosciuszko
(2228 m)
MOUNTAIN RANGES
Highest Peaks
Representational map. Not to scale.
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Exam Practice

- Practice your Geo skills with this map of Japan



ANSWERS:

Q1) What direction is Mount Fuji from Tokyo?
SW or WSW

Q2) True or False, Kyoto lies in a mountainous region?
True

Q3) How many km from Tokyo to Sapporo?
900 km (approx)

Q4) What is the body of water W of the Bonin Islands?
Philippine Sea

Q5) True or False, there are 1.6 miles to every 1 km?
False (it is the other way around)

Q6) What direction is Hiroshima from Miyazaki?
NNE

Q7) Describe the terrain in Kushiro?
low lying, flat, coastal terrain (low is the key word)